

DSA 210 – Introduction to Data Science

2025–2026 Spring Term | Final Report

# Momentum Collapse:

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## Analyzing Short-Term Performance Decline After Losing Break Points in Professional Tennis

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GitHub Repository: [github.com/nilugur/dsa210-term-project](https://github.com/nilugur/dsa210-term-project)

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*Additional presentation material: All code, data outputs, and the analysis notebook are available at [github.com/nilugur/dsa210-term-project](https://github.com/nilugur/dsa210-term-project)*

### 1. Motivation

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**Why this project?** *I wanted to find out whether the concept of “momentum collapse” in tennis — something every fan and commentator talks about — actually exists in the data. Tennis is a sport where a single point can shift the psychological balance of an entire match, yet no one had rigorously quantified what happens at the point level immediately after a critical miss.*

Tennis is a sport uniquely governed by pressure. Unlike team sports where individual errors are diffused across many players, a single professional tennis player must manage both physical performance and psychological composure simultaneously, often in front of thousands of spectators, against a single opponent. One of the most psychologically charged moments in a tennis match is the break point: a situation where the receiving player is one point away from breaking the opponent's serve and potentially shifting the momentum of an entire set.

While sports commentators frequently discuss momentum and psychological collapse in qualitative terms, the empirical literature on within-match performance fluctuation in tennis remains limited. Aggregate match statistics (e.g., first-serve percentage, winners, unforced errors) fail to capture sequential point-level dynamics. The core question motivating this project is:

*Do professional tennis players exhibit a measurable short-term performance decline in the points immediately following a lost break point opportunity?*

This question has real-world implications for coaching, performance analysis, match strategy, and sports psychology. If a quantifiable “momentum collapse” effect exists, it can inform how coaches prepare players for high-pressure situations, and how opponents strategize after surviving a break point. By applying data science methods to point-by-point match data, this project aims to move the conversation from anecdote to evidence.

## 2. Data Source

**Where did the data come from?** *Two publicly available datasets by Jeff Sackmann, downloaded from GitHub. How was it collected? The raw CSV files were downloaded directly from GitHub, loaded into Python using pandas, filtered to the relevant years (2015 and 2017), and merged into a single event-level dataset using a custom Python pipeline.*

### 2.1 Primary Datasets

Two publicly available datasets curated by Jeff Sackmann were used:

- Point-by-Point Data (tennis\_pointbypoint): This dataset records the sequence of every individual point played in professional ATP matches, encoded as character strings (e.g., 'S' = server wins, 'R' = returner wins, 'A' = ace, 'D' = double fault). The archive covers matches from 1991 to the present.
- ATP Match Data (tennis\_atp): This dataset contains match-level metadata including player rankings, surface type (Hard, Clay, Grass), tournament level, match scores, and player statistics for every ATP main-draw match.

### 2.2 Data Characteristics

After filtering to the target years (2015 and 2017) and merging the two datasets:

| Metric                   | Value      |
|--------------------------|------------|
| Total matched matches    | 4,342      |
| Total break point events | 36,267     |
| Years covered            | 2015, 2017 |

| Metric                  | Value                                                     |
|-------------------------|-----------------------------------------------------------|
| Point-by-point encoding | Character sequences per game                              |
| Surface types           | Hard, Clay, Grass, Carpet                                 |
| Tournament levels       | Grand Slams (G), Masters/Finals (M/F), Other (A, D, etc.) |

## 2.3 Data Enrichment

The point-by-point dataset alone does not contain player rankings, surface information, or tournament metadata. The tennis\_atp dataset was merged to enrich each event record with:

- Player and opponent ATP ranking at time of match
- Rank difference ( $\text{rank\_diff} = \text{player\_rank} - \text{opponent\_rank}$ )
- Underdog indicator ( $\text{is\_underdog} = 1$  if  $\text{player\_rank} > \text{opponent\_rank}$ )
- Surface type (Hard, Clay, Grass)
- Tournament pressure level (high = Grand Slams; medium\_high = Masters/Finals; lower = all others)
- Round of tournament

Merging was performed on the combination of year, winner name, loser name, and score string (after normalizing unicode characters and cleaning retirement/walkover markers).

## 3. Data Analysis

**Techniques used:** A custom Python data pipeline for preprocessing and merging; point-by-point parsing to extract break point events; feature engineering (PBPP, PD, rank features); exploratory data analysis (distributions, surface comparisons); one-sample t-tests and one-way ANOVA for hypothesis testing; and supervised machine learning (Logistic Regression, Decision Tree, Random Forest) for prediction.

### 3.1 Data Pipeline

A reproducible Python pipeline (dsa210\_tennis\_pipeline.py) was built to process raw data from disk and produce analysis-ready CSV files. The pipeline consists of five stages:

| Stage    | Description                                                                                                       |
|----------|-------------------------------------------------------------------------------------------------------------------|
| 1. Load  | Read PBP archive and current files; filter to target years; parse dates                                           |
| 2. Clean | Normalize player names (unicode, lowercase, punctuation removal); clean score strings                             |
| 3. Merge | Join PBP and ATP datasets on year + winner + loser + score; retain only matched records                           |
| 4. Parse | Iterate over each match's point-by-point string; identify break point situations; extract lost break point events |

| Stage                  | Description                                                                               |
|------------------------|-------------------------------------------------------------------------------------------|
| 5. Feature Engineering | Compute PBPP, PD, rank features, and surface/pressure categorical features for each event |

## 3.2 Feature Engineering

### Post-Break-Point Performance (PBPP)

The main engineered feature is PBPP<sub>k</sub> (Post-Break-Point Performance for window k), defined as:

$$\text{PBPP}_k = (\text{Points won by the player in the next } k \text{ points after losing a break point}) / k$$

Three window sizes were used:  $k = 3$ ,  $k = 6$ , and  $k = 12$ , to capture immediate, short-term, and medium-term effects.

### Performance Drop (PD)

To control for each player's overall skill level within the match, a baseline win rate was computed as the fraction of total points won by that player across the entire match. Performance Drop is then defined as:

$$\text{PD}_k = \text{PBPP}_k - \text{Baseline Win Rate}$$

A negative PD value indicates the player performed worse than their match average in the  $k$  points following a lost break point. This normalization ensures that weaker players (who naturally win fewer points) are not incorrectly attributed with a larger collapse effect.

### Break Point Identification

A break point is defined as any situation where the server has fewer points than the returner in a standard game, and the returner has at least 3 points (score of 0-40, 15-40, 30-40, or deuce with advantage to returner). The lost break point event is recorded when the server wins that point, denying the break. Tiebreaks are excluded from break point logic, as there are no break points in tiebreaks by definition.

## 3.3 Exploratory Data Analysis

### Dataset Overview

The event-level dataset contains 36,267 lost break point events drawn from 4,342 matched matches across two seasons (2015 and 2017).

| Feature          | Value   |
|------------------|---------|
| Total events (n) | 36,267  |
| Unique matches   | 4,342   |
| Mean PBPP (k=3)  | 0.432   |
| Mean PBPP (k=6)  | 0.502   |
| Mean PBPP (k=12) | 0.504   |
| Mean PD (k=3)    | -0.0794 |
| Mean PD (k=6)    | -0.0099 |
| Mean PD (k=12)   | -0.0068 |

## Surface Distribution

The dataset covers four surface types, with Hard court matches dominating:

| Surface | Event Count | Percentage |
|---------|-------------|------------|
| Hard    | 19,139      | 52.8%      |
| Clay    | 12,411      | 34.2%      |
| Grass   | 4,699       | 12.9%      |
| Carpet  | 18          | 0.1%       |

## Pressure Level Distribution

Tournament pressure was classified based on tournament level codes in the ATP dataset:

| Pressure Level | Definition                        | Event Count |
|----------------|-----------------------------------|-------------|
| High           | Grand Slams (G)                   | 9,824       |
| Medium-High    | Masters 1000 / Tour Finals (M, F) | 7,095       |
| Lower          | All other tournaments             | 19,348      |

## Key EDA Observations

- The distribution of  $PD_3$  is clearly left-shifted from zero, with a mean of  $-0.079$ , confirming that players win fewer points than their baseline in the 3 points immediately after losing a break point.
- The effect attenuates rapidly:  $PD_6$  ( $-0.010$ ) and  $PD_{12}$  ( $-0.007$ ) are much smaller in magnitude, suggesting the psychological impact fades quickly.
- Average PBPP by surface shows similar values across Hard, Clay, and Grass, indicating the effect is not strongly surface-dependent.

- PD distributions by surface overlap substantially, though ANOVA results show a marginally significant difference for  $k=3$ .

## 4. Hypothesis Testing

### 4.1 Hypotheses

For each window size  $k \in \{3, 6, 12\}$ , the following one-sided hypotheses were tested:

$H_0 : \mu(\text{PD}_k) = 0$  (no performance decline after a lost break point)

$H_1 : \mu(\text{PD}_k) < 0$  (players perform below baseline after a lost break point)

Additionally, one-way ANOVA tests were conducted to assess whether the mean PD differs significantly across surface types (Hard, Clay, Grass).

### 4.2 Statistical Tests and Results

One-sample t-tests (one-sided, lower tail) were used for the main hypothesis, and one-way ANOVA was used for the surface comparison. All tests used the full event dataset after removing rows with missing PD values (which occur when fewer than  $k$  points remain in the match).

| Test                               | n      | Mean PD | Statistic    | p-value (one-sided) |
|------------------------------------|--------|---------|--------------|---------------------|
| t-test: $\text{PD}_3 < 0$          | 35,585 | -0.0794 | $t = -51.69$ | $< 0.0001$ ***      |
| t-test: $\text{PD}_6 < 0$          | 35,157 | -0.0099 | $t = -9.11$  | $< 0.0001$ ***      |
| t-test: $\text{PD}_{12} < 0$       | 33,744 | -0.0068 | $t = -9.85$  | $< 0.0001$ ***      |
| ANOVA: $\text{PD}_3$ by surface    | 35,585 | -0.0794 | $F = 3.25$   | 0.021 *             |
| ANOVA: $\text{PD}_6$ by surface    | 35,157 | -0.0099 | $F = 1.93$   | 0.122 (ns)          |
| ANOVA: $\text{PD}_{12}$ by surface | 33,744 | -0.0068 | $F = 1.18$   | 0.315 (ns)          |

Significance codes: \*\*\*  $p < 0.001$ , \*  $p < 0.05$ , ns = not significant

### 4.3 Interpretation

- All three one-sample t-tests reject the null hypothesis at  $p < 0.001$ . Players demonstrably perform below their baseline win rate in the points immediately following a lost break point.
- The strongest effect is at  $k = 3$  (mean PD = -0.079,  $t = -51.69$ ), indicating the decline is most pronounced in the first 3 points after the event. This is consistent with a short-term psychological shock.
- The effect attenuates substantially by  $k = 6$  (-0.010) and  $k = 12$  (-0.007), suggesting players recover relatively quickly. The momentum collapse is real but transient.

- The ANOVA for  $k = 3$  is significant ( $p = 0.021$ ), meaning surface type does account for a small but statistically detectable variation in the immediate collapse magnitude. However, surface differences are not significant at  $k = 6$  or  $k = 12$ , and the effect sizes are small.
- The massive t-statistics ( $|t| > 9$  for all  $k$ ) combined with the large sample sizes confirm that this is not a chance finding — the performance drop is a genuine, consistent phenomenon across professional tennis.

## 5. Machine Learning

### 5.1 Problem Formulation

Building on the hypothesis testing results, supervised classification models were trained to predict whether a player will experience a performance drop ( $PD_3 < 0$ ) after losing a break point. The target variable is:

```
collapse = 1   if  $PD_3 < 0$    (player performs below baseline)   |
collapse = 0   otherwise
```

$k = 3$  was chosen as the window because it showed the strongest and most statistically significant effect in hypothesis testing. Predicting individual-level collapse extends the population-level inference from hypothesis testing to actionable, event-level prediction.

### 5.2 Features

All features used as model inputs are available before the break point event occurs — ensuring no data leakage:

| Feature Type                  | Features                                                                                          |
|-------------------------------|---------------------------------------------------------------------------------------------------|
| Numerical                     | player_rank, opponent_rank, rank_diff, baseline_win_rate, is_underdog, set_no, game_no            |
| Categorical (one-hot encoded) | surface (Hard/Clay/Grass), pressure_level (high/medium_high/lower), round (R32/R16/QF/SF/F, etc.) |

### 5.3 Models Trained

| Model               | Key Hyperparameters                                                   |
|---------------------|-----------------------------------------------------------------------|
| Logistic Regression | $C = 1.0$ , class_weight = balanced, StandardScaler applied           |
| Decision Tree       | max_depth = 5, min_samples_leaf = 50, class_weight = balanced         |
| Random Forest       | n_estimators = 200, max_depth = 10, class_weight = balanced_subsample |

Evaluation method: 80/20 stratified train/test split with 5-fold cross-validation. Metrics reported: Accuracy, Precision, Recall, F1-score, and ROC-AUC.

Class imbalance (more collapses than non-collapses, given the negative mean PD) was handled via `class_weight = balanced` across all models.

### 5.4 Model Performance Results

The table below shows the full evaluation results from the test set and 5-fold cross-validation, drawn directly from the analysis notebook:

| Model               | Accuracy | Precision | Recall | F1     | ROC-AUC (test) | CV ROC-AUC      |
|---------------------|----------|-----------|--------|--------|----------------|-----------------|
| Logistic Regression | 0.5791   | 0.7146    | 0.5838 | 0.6426 | 0.6009         | 0.6018 ± 0.0083 |
| Decision Tree       | 0.5717   | 0.7151    | 0.5638 | 0.6305 | 0.6006         | 0.5944 ± 0.0066 |
| Random Forest       | 0.6051   | 0.7158    | 0.6480 | 0.6802 | 0.6234         | 0.6152 ± 0.0070 |

**Best F1 Score:** Random Forest (0.6802)    **Best ROC-AUC:** Random Forest (0.6234 test / 0.6152 ± 0.0070 CV)

### 5.5 Key Findings

- `baseline_win_rate` is the most predictive feature across all models. Players who win fewer points overall are more vulnerable to collapse, which is consistent with the hypothesis test findings.
- `rank_diff` and `player_rank` also contribute significantly — higher-ranked (lower numerical rank) players show more resilience after a lost break point, suggesting that elite players have stronger mental recovery mechanisms.
- `set_no` contributes modest predictive power — collapse effects appear to intensify in later sets, consistent with physical and psychological fatigue.
- Random Forest achieved the best overall performance across Accuracy, F1, and ROC-AUC metrics, benefiting from its ability to capture non-linear interactions between features.
- Moderate AUC values (typical for behavioral prediction tasks) indicate that while structural features (ranking, surface, round) carry genuine predictive signal, there is an irreducible random and psychological component that pre-match observable features alone cannot fully capture.
- The ML results extend the hypothesis tests: moving from population-level statistical inference to individual-level prediction of which players and contexts are most likely to experience collapse.

## 6. Findings

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**Key finding:** *Yes — professional tennis players do experience a statistically significant performance decline immediately after losing a break point. The effect is largest in the first 3 points (mean drop of ~8 percentage points below baseline) and fades by 12 points, confirming it is a short-term psychological phenomenon.*

## 6.1 Primary Finding: Momentum Collapse is Real

The central finding of this project is that professional tennis players do experience a statistically significant and practically meaningful performance decline in the points immediately following a lost break point. Across a dataset of 36,267 events and 4,342 matches:

- Players win only 43.2% of the next 3 points after losing a break point, compared to their match baseline win rate of approximately 51–54%.
- The mean Performance Drop at  $k = 3$  is  $-0.079$ , representing a nearly 8 percentage point decline below the player's own baseline.
- This effect is highly statistically significant ( $t = -51.69$ ,  $p < 0.0001$ ) and observed consistently across both years, all surfaces, and all tournament levels.

## 6.2 The Effect is Short-Lived

The momentum collapse is transient:

- $PD_3 = -0.079$  (strongest, immediate effect)
- $PD_6 = -0.010$  (rapid attenuation by 6 points)
- $PD_{12} = -0.007$  (near-zero by 12 points)

This pattern is consistent with a psychological shock model: the negative experience of missing a break point opportunity causes a momentary loss of focus or confidence, but players recover within a few games. The finding supports the narrative of 'momentum' in tennis as a real but ephemeral phenomenon.

## 6.3 Surface Effects are Minor

Surface type accounts for a statistically significant portion of variance in  $PD_3$  (ANOVA  $p = 0.021$ ), but not for  $PD_6$  or  $PD_{12}$ . The effect sizes across surfaces are small, suggesting that the collapse phenomenon is universal across playing conditions and not substantially moderated by surface.

## 6.4 Player Quality Predicts Recovery

Machine learning models confirm that higher-ranked players (better players) are less likely to experience a collapse after a lost break point. The baseline win rate is the single strongest predictor — players who win more points overall tend to recover faster. This aligns with the sports psychology literature on mental toughness and experience under pressure.

# 7. Limitations and Future Work

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**What could be done better?** *The dataset could be expanded to more seasons, player-level random effects could be modeled, and the analysis could be extended to the WTA tour and to regression (predicting PD magnitude rather than binary collapse).*

## 7.1 Limitations

- Dataset scope: The analysis covers 2015 and 2017 ATP matches only. Expanding to all available seasons (1991–2024) would substantially increase statistical power and enable trend analysis over time.
- Missing data: PBPP values are missing for events where fewer than  $k$  points remain in the match. These were removed, which could introduce selection bias (events near the end of matches are excluded).
- Break point definition: This study uses only “lost break point” events (server wins the break point). It does not analyze the parallel situation for the server (saved break point), which may have a different psychological profile.
- No player-level fixed effects: The analysis treats all players identically. Player-specific baseline effects (e.g., Djokovic vs. an average player) are only partially captured through the baseline win rate. A mixed-effects model with player random effects would be more rigorous.
- Baseline win rate confound: Using the full-match baseline win rate as the normalizer may be contaminated by the collapse events themselves, slightly biasing PD estimates toward zero.
- Tiebreak exclusion: Tiebreak games are excluded from break point analysis. High-stakes tiebreak points could exhibit similar or stronger momentum effects.
- Causal inference: This study establishes correlation, not causation. The observed PD could reflect tactical adjustments by the opponent (who just “survived” a break point) rather than purely psychological effects on the player who missed.

## 7.2 Future Work

- Regression approach: Train regression models to predict the continuous PD value rather than a binary collapse indicator, capturing the magnitude of the decline.
- Temporal trends: Investigate whether the collapse effect has changed over time (e.g., is modern tennis more or less susceptible to momentum collapse?).
- Women's tour (WTA): Replicate the analysis on WTA data to assess whether the effect differs by gender.
- Player-level analysis: Identify which individual players are most and least susceptible to momentum collapse, enabling personalized coaching recommendations.
- Opponent perspective: Analyze the symmetric situation — does the player who saves the break point benefit from a positive momentum boost?
- Live prediction: Adapt the model for real-time use during matches to flag high-collapse-risk situations for coaches in the tunnel.

## 8. AI Usage Disclosure

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In accordance with the DSA 210 academic integrity policy, all use of AI tools is explicitly documented below. AI assistance was used only as a supporting tool; the research idea, dataset selection, analytical decisions, and conclusions are entirely the student's own work.

### 8.1 What Was Done Independently (No AI Involved)

- Topic selection: The idea of studying momentum collapse in tennis after lost break points was conceived independently by the student.
- Dataset selection: The two Jeff Sackmann datasets (tennis\_pointbypoint, tennis\_atp) were identified and chosen by the student.
- Proposal writing: The project proposal document was written by the student.
- Final report writing: The content, structure, and arguments in this final report were determined by the student.
- Analysis decisions: All methodological choices — how to define break points, which window sizes ( $k = 3, 6, 12$ ) to use, which statistical tests to apply, which ML models to train — were made by the student.

## 8.2 Where AI Assistance Was Used

| Area                 | AI Tool              | How It Was Used                                                                                                                                                                |
|----------------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Proposal document    | Gemini               | Grammar and readability corrections on student-written text. The ideas and content were the student's own.                                                                     |
| Final report         | Claude (Anthropic)   | Grammar, phrasing, and intelligibility improvements on student-written sections. Structure and content reflect the student's own analysis.                                     |
| Data pipeline & code | AI coding assistants | Debugging help and syntax suggestions during development of dsa210_tennis_pipeline.py and analysis.ipynb. All logic and algorithmic design decisions were made by the student. |
| Some calculations    | AI coding assistants | Assistance verifying certain numerical computations and statistical formulas. All results were checked against the student's own outputs.                                      |

## 8.3 Summary Statement

*AI tools were used exclusively to improve the quality of expression (grammar, clarity) and to assist with technical coding tasks. The research topic, dataset, analytical framework, hypothesis formulation, interpretation of results, and all conclusions presented in this report are the original work of Nil Uğur.*

## References

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